GG-C-001072A(GSA-FSS)
October 3, 1978
SUPERSEDING
GG-C-001072
April 7, 1967

INTERIM FEDERAL SPECIFICATION

CLOCKS, WALL (QUARTZ CRYSTAL - BATTERY OPERATED)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers quartz crystal wall clocks powered by one or two 1.5 volt C or D size dry cells, suitable for use in Government offices and/or activities. The clocks shall be capable of keeping accurate time within plus or minus 10 seconds per month (one second gain or loss in 72 hours).

1.2 Classification. Wall clocks covered by this specification shall be of the following types, classes, and styles as specified (6.1):

Type I - 8 inch dial
Type II - 11-1/2 inch dial
Class A - 12 hour clock
Class B - 24 hour clock
Style 1 - White dial, black figures
Style 2 - Black dial, white figures

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 6645

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of Military Specifications and Standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 General. Clocks supplied under this specification shall be completely assembled and ready for operation, requiring only the insertion of the proper dry cell or cells (1.5 volt, C or D size). It shall not be necessary to perform periodic movement adjustments to obtain the timing accuracy specified herein.

3.1.1 The clock unit shall be equipped with a concealed hanging device, suitable for hanging the clock flush with the wall.

3.2 Cases. Cases shall be of the surface type, fabricated and finished of steel or molded plastic material in accordance with the manufacturer's standard commercial practice. Unless otherwise specified (6.1), metal cases shall be finished in bronze, brown, or white enamel which will not be easily chipped or cracked, while the molded plastic cases shall be supplied in bronze, brown, or white with a standard mold finish. All cases shall have dial covers of either commercial clear glass or commercial nonyellowing clear plastic. The dial cover shall be securely positioned to prevent accidental separation of the cover from the case.

3.2.1 Dial. Unless otherwise specified (see 3.2.1.4), twelve hour dials shall be furnished. For type I clocks, the diameter of the dial (sight opening) shall be 8 inches, plus or minus three-eights (3/8) inch, and for type II clocks, the diameter of the dial (sight opening) shall be 11-1/2 inches, plus or minus one-half (1/2) inch. Dials shall have arabic numerals and minute marks and shall be treated to prevent discoloration in accordance with commercial practice. When the dial is of material other than metal, it shall be securely fastened to a metal or other base in such a manner as to prevent warping, buckling or any irregularities in appearance or positive fit.

3.2.1.1 Dials shall have a second hand in addition to an hour and a minute hand. The second hand shall be finished in some distinguishing color.

3.2.1.2 The minute hand for type I and type II clocks shall extend a minimum of three and one-half (3-1/2) inches from the center of the clock. The hour hand for both type I and type II clocks shall be no more than 85 percent of the length of the minute hand.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

3.2.1.3 The arabic numerals and the marks indicating minutes shall be of a height, length, and width reflecting the manufacturer's standard commercial practice.

3.2.1.4 When specified in the invitiation for bids (see 6.1), dials shall register 24 hours in one revolution of the hour hand, and 1 hour in one revolution of the minute hand. The numerals indicating the hours "00" (or 24), "4", "8", "12", "16", and "20" shall be dimensionally higher than the remaining numerals in accordance with the manufacturer's standard practice.

3.3 Detail of components.

3.3.1 Movement. Clocks offered under this specification shall be factory adjusted to meet the timing accuracy specified herin.

3.3.2 Reset. The device for resetting the clock shall be a projection from the bottom or back of the case.

3.4 Identification. The manufacturer's name, model number, serial or production run identification number, shall be marked on the back of the clock.

3.4.1 Instructions. The manufacturer shall provide whatever operational and/or maintenance instructions/recommendations are deemed appropriate.

3.5 Performance.

3.5.1 Accuracy. Clocks furnished under this specification shall not gain more than one (1) second in 72 hours nor lose more than one (1) second in 72 hours when tested as specified in 4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105 as specified in table I below.

TABLE I

Examination	Inspection Level	AQL (Percent Defective)
Major	S-4	4.0
Minor	S-4	6.5
Test	S-3	4.0

4.3 Examination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

4.3.1 Classification of defects. For inspection purposes, defects shall be classified as follows:

Examine	Defect	Major	Minor
Case	Material used other than specified	X	
	Surface not smooth	X	
	Improper fit of case to dial and cover glass	X	
Clock-General	Not design specified	X	
	Case improperly finished	X	
	Burrs or undersirable projecting ridges at the dividing line of mold		X
	Any component cracked, fractured, broken or damaged	X	
	Clock will not run	X	
	Device for resetting clock not as specified	X	
	Identification, missing, illegible, not as specified		X
Dial Cover	Cover broken	X	
	Poor quality, not clear, blisters, bubbles, scratch or scratches, coarse or open seeds, heavy seeds or feathers, other defects which distort or impair the vision		X
Dial	Dial not as specified	X	
	Any dial marking missing, illegible, incomplete, incorrect, or wrong material used to apply	X	
	Dial surface scratched, abraded, or otherwise damaged	X	
Hands	Any hand not securely mounted	X	
	Any hand bent, malformed, or otherwise damaged	X	
	Hand rubs against glass or dial, or hand obstructs movements of another hand	X	
	Hands not as specified; color, size, etc.	X	

4.4 Tests.

4.4.1 Accuracy. Clocks to be tested shall be placed in a vertical position in a room maintained at the specified temperature for at least 8 hours prior to making the test. Clocks shall be powered by a regulated d.c. power supply for the test. The accuracy of the standard timing device shall be traceable to the National Bureau of Standards time signals transmitted by WWV or WWVH. (NBS Publication 236, "NBS Frequency and Time Broadcast Services".) These signals are available by telephone (303) 499-7111). Each sample clock shall be tested in accordance with the test conditions for test 1 and test 2 of table II. Sample clocks which do not meet the accuracy requirements of 3.5 shall be considered to have failed.

Source: https://assist.dla.mil -- Downloaded: 202640-26T11:53Z
Check the source to verify that this is the currentversion before use.

TABLE II

Test No.	Temperature	Test Voltages One-cell clocks	Two-cell clocks
1	75° ± 7-1/2°F	1.50 ± 1%	3.00 ± 1%
2	75° ± 7-1/2°F	1.30 ± 1%	2.00 ± 1%

4.4.1.1 Measurement of accuracy. Instrumentation capable of measuring an accuracy of 0.1 second or better shall be used to determine the time of the sample clocks relative to the time signals from NBS(WWV). A dual-channel stripchart recorder that will respond to the WWV signal shall be used for the time comparison. Channel #1 records the WWV signal and the standard time is noted at each 1 minute beep. (The WWV electrical signal may be obtained across the speaker of a desk telephone amplifier or a radio speaker tuned to the WWV signal. A local standard telephone or radio time signal, or secondary standard clock may be used if it is traceable to WWV time signals.) Channel #2 records the movement of the second hand as it passes the 45 second position of the clock. (The 45 second position is used to eliminate the backlash of the second hand. The 15 second position may be used in place of the 45 second position.) The signal for a sweep-second-hand movement shall be obtained with a photoelectric switch. The second hand interrupts a light beam at the 45 second position of the clock. The signal for a stepper-second-hand movement shall be obtained by sensing the pulsed current drawn by the clock. (See Figure 1) (The recorder's event marker may be needed to locate the current pulse at the 45 second position of the clock.)

Initial and final readings for channels #1 and #2 shall be made to determine the test clock's accuracy. The initial reading is taken by recording the Clock's 45 second position point on channel #2 relative to the nearest 1 minute standard time signal beep. (Usually a point on the leading edge of the beep signal is used.) (Figure 2) The recorder shall be turned off until the final reading is taken, at the end of 72 hours.

The final reading is repeated in the same manner as the initial reading using the 45 second position. After obtaining the final reading, the initial and final charts are compared. By comparing the difference, x, (in seconds) between the 45 second position of the clock (channel #2) and the nearest standard minute beep (channel #1), (see Figure 2) to the difference of the final readings, y, (in seconds) (see Figure 3), the accuracy of the test clock can be determined using the following formula: $t = x - (y)$. If y value of the final reading is located on the other side of the standard minute beep from the x reading, relative to the initial reading, then the sign of the y value changes. (see Figure 4).

4.5 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of Section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or commercial, as specified (see 6.1).

5.1.1 Level A. Each clock shall be packaged in a close-fitting polyethylene bag constructed from material a minimum of 1.5 mils thick. The bag shall have the excess air removed to the practicable minimum and shall be heat sealed closed. The clock shall be further packaged in a close-fitting box conforming to PPP-B-636, class weather resistant, style optional. Closure and waterproof sealing of the box shall be in accordance with the appendix to PPP-B-636.

5.1.2 Commercial. The clocks shall be individually packaged in accordance with normal commercial practice. The complete package shall be designed to protect the clocks against damage during shipment, handling, and storage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.1).

5.2.1 Level A. Clocks of like description, packaged as specified in 5.1.1, shall be packed in quantities as specified in the contract or order in a close-fitting box conforming to PPP-B-636, class weather resistant, style optional. Closure and waterproof sealing of the boxes shall be in accordance with the appendix to PPP-B-636. Alternatively, cleated plywood, wire bound or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The edges of the barrier material shall be sealed with waterproof tape or adhesive.

5.2.2 Commercial. The clocks of like description, packaged as specified in 5.1.2, shall be packed in fiberboard boxes to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking. Packages and shipping containers shall be marked as specified in the contract or order (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, class, and style desired (see 1.2.1).
(c) Finish of metal case if other than specified (see 3.2).
(d) Level of packaging, packing, and marking required (see 5.1 and 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

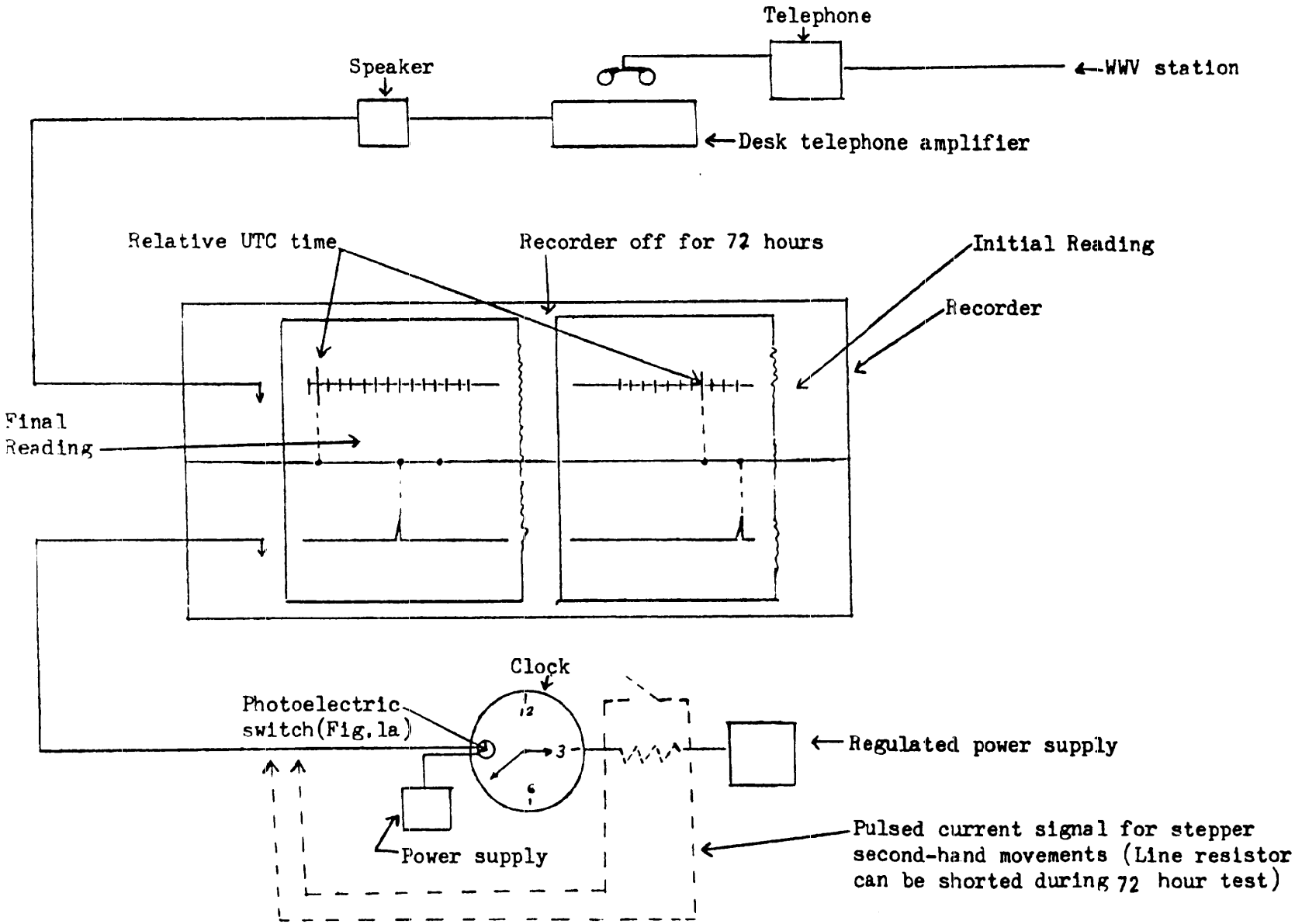
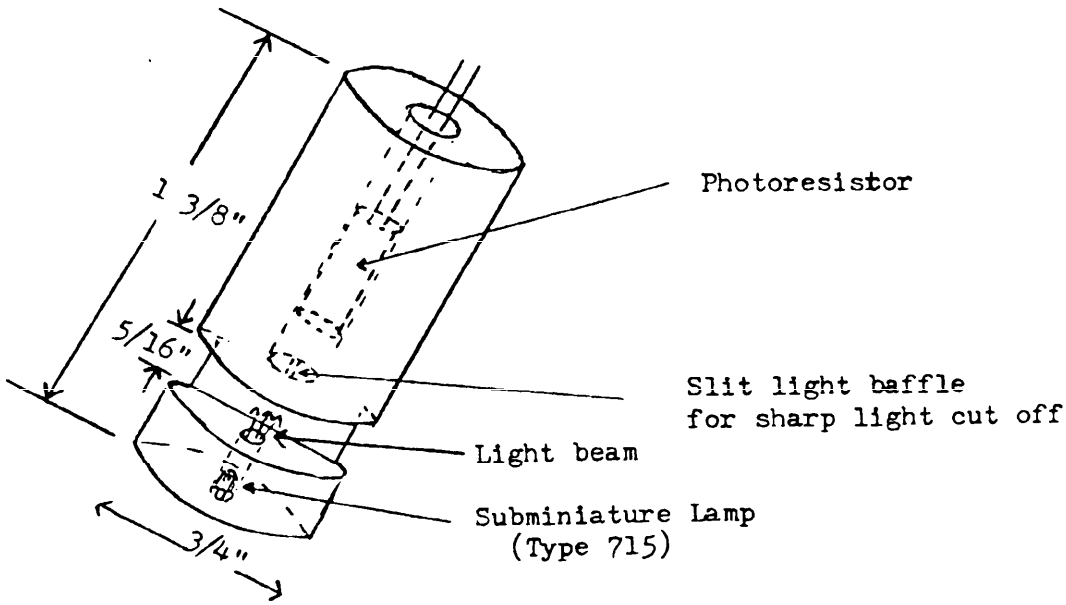


Figure 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.



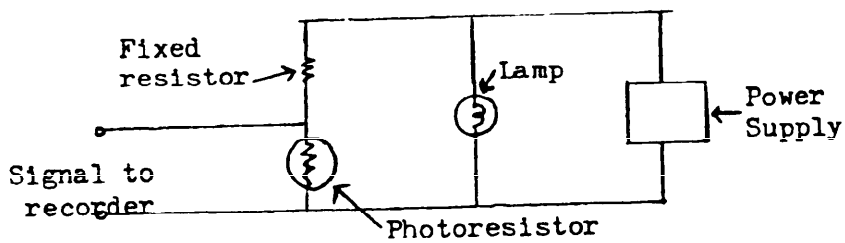


PHOTO-ELECTRIC SWITCH

FIGURE 1a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

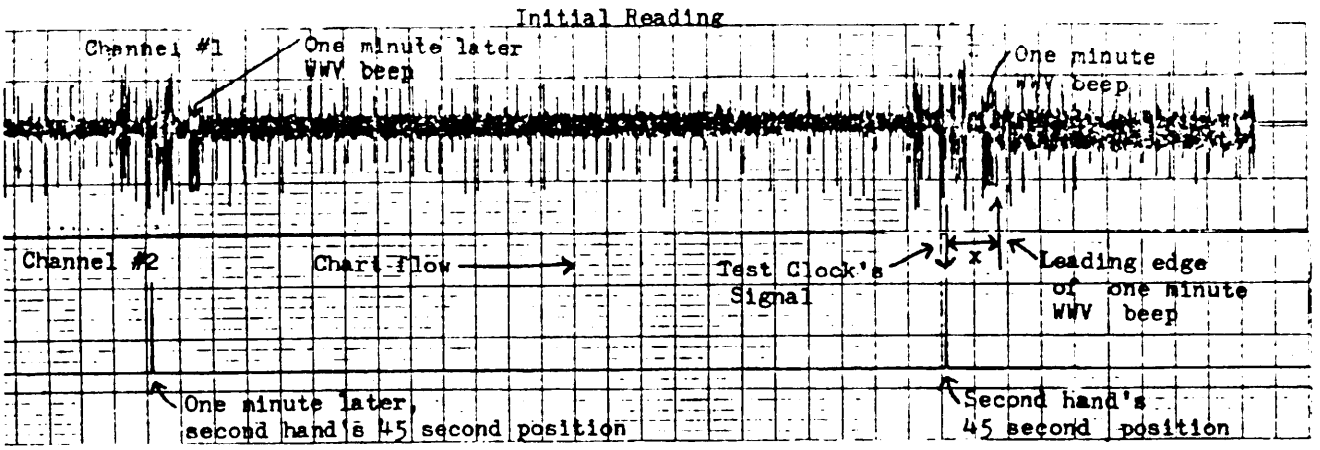


Figure 2

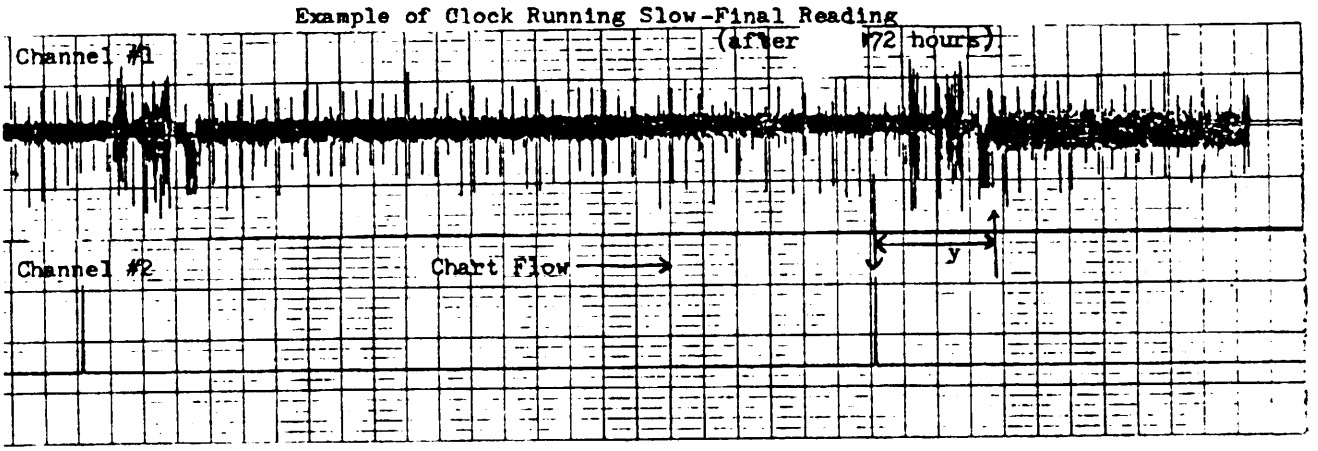


Figure 3

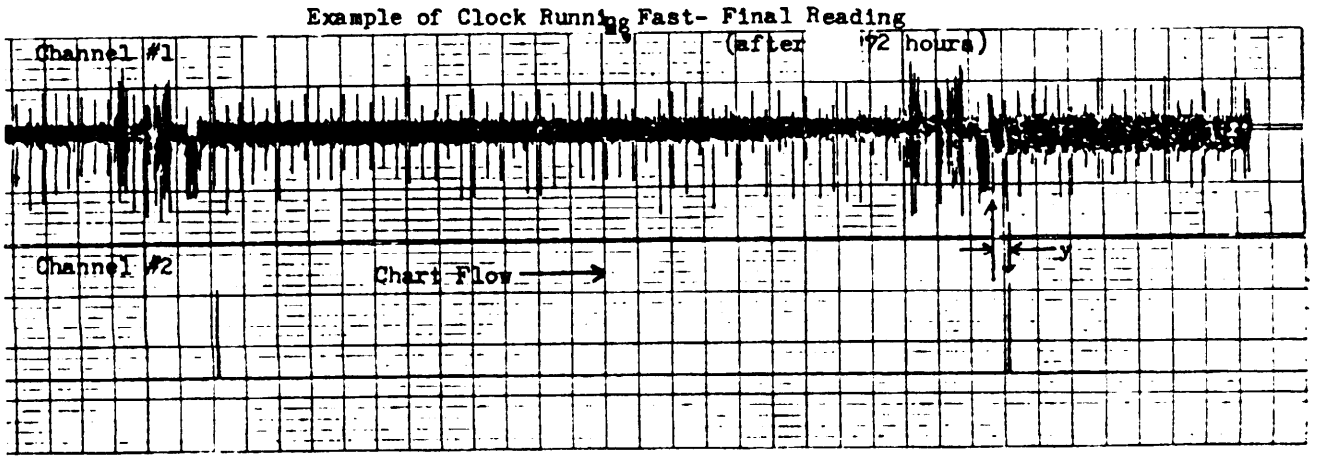


Figure 4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:53Z
Check the source to verify that this is the current version before use.